

Tue

Name

- For the ODE $y' = Ay$ where $A = \begin{pmatrix} -3 & -2 \\ -2 & -3 \end{pmatrix}$

`In[ ]:= MatrixForm[Eigensystem[ $\begin{pmatrix} -3 & -2 \\ -2 & -3 \end{pmatrix}$ ]]`

`Out[ ]:=MatrixForm=`

$$\begin{pmatrix} -5 & -1 \\ \{1, 1\} & \{-1, 1\} \end{pmatrix}$$

- Write down the gen sol and the important part as $t \rightarrow \infty$.

$$y(t) = c_1 e^{-5t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \approx c_2 e^{-t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

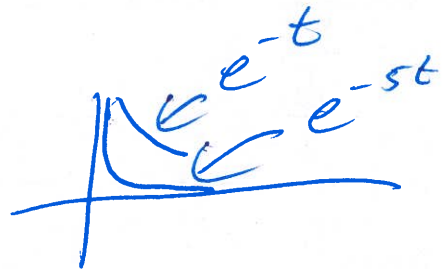
- Write the gen sol of $y' = Ay$ with $A = \begin{pmatrix} -3 & 2 \\ -2 & -3 \end{pmatrix}$. Describe the solution as $t \rightarrow \infty$.

$$y(t) =$$

`In[ ]:= MatrixForm[Eigensystem[ $\begin{pmatrix} -3 & 2 \\ -2 & -3 \end{pmatrix}$ ]]`

`Out[ ]:=MatrixForm=`

$$\begin{pmatrix} -3 + 2i & -3 - 2i \\ \{-i, 1\} & \{i, 1\} \end{pmatrix}$$



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$$a + bi = -2 + 3i$$

$$a = -2$$

$$b = 3$$

~~$$p + qi = \{1 - i, -i, 1\}$$~~

$$p = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, q = \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$$

■ Write down the gen solution of ODE $y' = A y$ with A below.

Give the most important part as $t \rightarrow \infty$.

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In[ ]:= MatrixForm[Eigensystem[ $\begin{pmatrix} 5 & -10 & -4 \\ 4 & -6 & -1 \\ 2 & -5 & -4 \end{pmatrix}$ ]]
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Out[]:= MatrixForm=

$$\begin{pmatrix} -2 + 3i & -2 - 3i & -1 \\ \{1 - i, -i, 1\} & \{1 + i, i, 1\} & \{-1, -1, 1\} \end{pmatrix}$$

$$y_1(t) = e^{at} (\cos(bt)p - \sin(bt)q)$$

$$y_2(t) = e^{at} (\cos(bt)q + \sin(bt)p)$$

$$y(t) = y_1(t) + c_1 y_2(t) + c_3 e^{-t} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \approx c_3 e^{-t} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

2. Write down the gen solution of ODE $y' = A y$ with A below.

Give the most important part as $t \rightarrow \infty$.

$$y(t) = \quad \approx \quad$$

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In[ ]:= Eigensystem[ $\begin{pmatrix} -3 & 4 & 0 \\ -4 & 3 & 2 \\ 0 & 2 & -1 \end{pmatrix}$ ] // MatrixForm
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Out[]:= MatrixForm=

$$\begin{pmatrix} -1 + 2i & -1 - 2i & 1 \\ \{1 + i, i, 1\} & \{1 - i, -i, 1\} & \{1, 1, 1\} \end{pmatrix}$$